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Ionic liquids as alternative stabilizers for lipid nanoparticles used to deliver mRNA

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The success of COVID-19 mRNA vaccines has drawn significant interest in developing lipid nanoparticles (LNPs) for gene delivery. Traditional LNP formulations employing polyethylene glycol-based stabilizers have been associated with potential limitations such as immunogenicity and accelerated clearance upon repeated administration. This study introduces LNPs incorporating choline-based ionic liquids (ILs) instead of PEGylated stabilizers for mRNA delivery, particularly to transfect alveolar macrophages. We systematically investigated LNPs incorporated with three choline-based ILs (choline hexanoate, choline aspartate and choline glutamate) including their physicochemical properties, stability and *in vitro* biological performance. It was found that ILs with amino acid anions, such as aspartate and glutamate, could lead to relatively stable LNP formulations in the absence of PEGylated stabilizers. These IL-incorporated LNPs exhibited notably improved intracellular mRNA transfection in alveolar macrophages compared to copolymer

F127-stabilized LNPs. We revealed acidification-induced structural transitions into more ordered inverse lipid mesophases of IL-incorporated LNPs, which may facilitate endosomal escape and efficient mRNA transfection. This LNP property-biological performance relationship study provides valuable insights into designing next-generation nanocarriers for gene delivery. The incorporation of ILs offers a promising avenue to modify the property and performance of LNPs for nanomedicine and mRNA vaccines.

This article is part of the discussion meeting issue 'Ionic liquids and the future of soft materials'.

1. Introduction

The potential and advancement of mRNA-based therapeutics and vaccines, highlighted by the rapid deployment and success of COVID-19 mRNA vaccines, underscore lipid nanoparticles (LNPs) as an excellent nanocarrier for gene therapeutics [1–3]. LNPs have the ability to encapsulate and deliver nucleic acids intracellularly via cellular uptake and endosomal escape [4,5]. Currently, typical LNPs used in RNA vaccines and medicine, including the COVID-19 mRNA vaccines (Pfizer-BioNTech and Moderna mRNA vaccines) and Onpattro®, are based on four components, an ionizable amino-lipid, a phospholipid (such as 1,2-distearoyl-sn-glycero-3-phosphocholine, DSPC), cholesterol and a polyethylene glycol (PEG)-lipid [6,7]. While significant efforts have been devoted to developing ionizable lipids, structural helper lipids and cholesterol, PEGylated materials are currently the most widely used stabilizers in commercial genetic medicine owing to their ability to provide steric stabilization, reduce aggregation and prolong circulation time by imparting a 'stealth' characteristic to the nanoparticles [8]. However, the broad utilization of PEGylated materials has raised concerns regarding accelerated blood clearance and immunogenic responses upon repeated administration, potentially limiting their long-term therapeutic applications [9]. Several groups have reported that mRNA vaccines, especially mRNA-1273 from Moderna, can enhance PEG-specific antibodies in humans [10–14]. For example, Ju *et al.* reported the boost of anti-PEG IgG and IgM in humans by the SARS-CoV-2 mRNA vaccine, which correlates with the increased systemic reactogenicity and LNP-leucocyte association in human blood [10]. In a follow-up study by Kent *et al.* it was further demonstrated that the induction of anti-PEG antibodies following repeated exposure to PEGylated mRNA vaccines resulted in reduced nanoparticle circulation time.[15] Similarly, Carreno *et al.* [14] and Kozma *et al.* [11] also reported the consistent significant increase in anti-PEG antibodies after mRNA-1273 (Moderna COVID-19 vaccine) vaccination. These findings underscore the need to develop PEG alternative stabilizers with improved biocompatibility and lower immunogenicity for next-generation LNP systems.

To address the emerging health concerns of PEG, expanding the toolbox of stabilizers for LNPs is highly desirable in the field of genetic nanomedicine. Novel stabilizers, such as polysarcosine lipids [16,17], amphiphilic polymers (such as those based on poly(N-vinyl pyrrolidone) [18], poly(2-methyl-2-oxazoline) [19,20], zwitterionic polymer–lipid conjugate) [21] and ionic liquids (ILs) [22–25], have been developed and show potential to enhance the stability, efficacy and safety profiles of LNP formulations for RNA delivery. Among them, the third-generation ILs developed since 2000 have emerged as a promising candidate class owing to their biodegradability and biocompatibility [26–28]. ILs have been used to modulate protein coronas, enhance drug solubility, disrupt membranes and act as functional motifs in responsive delivery systems in the nanomedicine field [24,29–31]. Choline-based ILs have shown reduced toxicity and enhanced compatibility with biological systems [32–34], making them suitable for biomedical application investigation. Recently, Khare *et al.* reported that choline trans-2-hexenoate-coated/incorporated LNPs significantly enhanced small interfering RNA (siRNA) delivery to mouse brain endothelial cells and motor neurons [22]. By engineering LNPs with

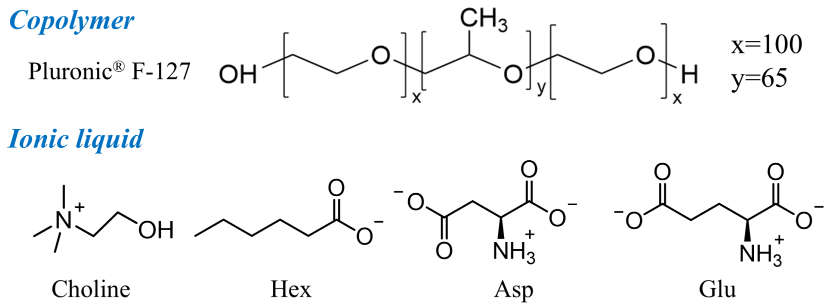

Figure 1. Molecular structures of the copolymer (F127) and choline-based ILs with anions of hexanoate (Hex), aspartate (Asp) and glutamate (Glu).

ILs, increased cellular uptake, enhanced blood-brain barrier permeability and improved siRNA transfection efficiency were achieved. This approach leverages the ability of ILs to interact with lipid membranes, facilitating more efficient nucleic acid delivery across biological barriers, which highlights the potential of IL-functionalized LNPs for gene therapy.

The development of mRNA-LNPs to reprogram M2 cancer-supportive macrophages into an M1 cancer-killing phenotype highlights the significant potential to transfect macrophages via gene therapy for the treatment of lung diseases and other pathologies [35]. In our previous work, we have reported that LNPs composed of SM-102, monoolein (MO) and cholesterol stabilized by block copolymer F127 are a feasible LNP system for transfecting macrophages [36,37]. F127 is a polyethylene oxide (PEO)-polypropylene oxide block copolymer and is therefore a PEGylated material. Herein, we engineered IL-incorporated LNPs of similar compositions, replacing the F127 polymer, for mRNA delivery to transfect alveolar macrophages, which are critical for pulmonary host defence [38,39]. This study aims to elucidate the effect of choline-based IL incorporation on the physicochemical properties and *in vitro* biological performance of LNPs for mRNA delivery to macrophages. We have systematically evaluated the effects of incorporating three choline-based ILs: choline hexanoate (Cho Hex), choline aspartate (Cho Asp) and choline glutamate (Cho Glu), on particle size, dispersity, surface charge, stability, pH-dependent lyotropic liquid crystalline lipid mesophase transitions, cell association and transfection efficiency of a reporter mRNA in a macrophage cell line. We have observed an improvement in mRNA transfection efficiency to approximately 85% for all IL-incorporated LNPs, which is significantly higher than the F127-stabilized LNPs (approx. 50%). In addition, the physicochemical property-biological function relationship of these IL-incorporated LNPs was further deciphered by connecting their pH-dependent lipid mesophase evolution and their transfection efficiency. This knowledge is valuable for optimizing LNP formulations stabilized with PEG alternatives and enhancing the therapeutic potential of mRNA therapeutics and vaccines.

2. Results and discussion

(a) Formation of ionic liquid-incorporated lipid nanoparticles and mRNA@LNPs

The IL-incorporated LNPs were generated by mixing lipid solvent solution with IL-containing (10 wt% IL related to the total lipid amount) citrate buffer (10 mM, pH 3) using a microfluidic platform and then removing solvent residue and free ILs in the bulk solution by dialysis in Milli-Q water. The molecular structures of the copolymer F127 and the choline-based IL used in this study, including choline hexanoate (Cho Hex), choline aspartate (Cho Asp) and choline glutamate (Cho Glu), are shown in figure 1. The LNPs were formulated using a lipid methanol solution with a fixed SM-102/MO/Cholesterol molar ratio of 50 : 30 : 20; the sample names

Table 1. Name and composition of the IL-incorporated LNPs used in the study.

sample name	lipid molar ratio (SM-102/MO/cholesterol, with a total lipid mass concentration of 15 mg ml^{-1})	IL/F127 (0.5 mg ml^{-1})
Hex	50/30/20	Cho Hex
Asp	50/30/20	Cho Asp
Glu	50/30/20	Cho Glu
F127	50/30/20	Copolymer F127

are listed in table 1. Our previously developed copolymer F127-stabilized LNPs of the same composition [36,37], which have been used to demonstrate the capacity to transfect murine alveolar macrophage MH-S cells, are included for comparison. Cho Hex has been chosen considering Khare *et al.*'s work, which observed enhanced efficiency of choline trans-2-hexenoate-coated/incorporated LNPs for siRNA delivery [22]. The choline-amino acids ILs, Cho Asp and Cho Glu, were selected based on their low cell toxicity in four cell lines (HDF, A431, AGS and HeLa cells) after screening 13 choline-based ILs in our recent work [40].

To investigate the incorporation of choline-based ILs (Hex, Asp and Glu) into LNP formulations to replace the copolymer F127, we first evaluated particle size (Z_{ave}) and polydispersity index (PDI) of IL-incorporated LNPs compared to F127-stabilized controls using dynamic light scattering (DLS) measurements, as shown in figure 2*a–c*. Among the formulations, all IL-incorporated LNPs exhibited a mean particle size of approximately 150 nm, Hex (149 ± 1 nm), Asp (130 ± 5 nm) and Glu (148 ± 6 nm), which are obviously larger than F127 LNPs (77 ± 3 nm). This size difference can possibly be attributed to the particle formation mechanisms. As a non-ionic triblock copolymer, F127 stabilizes nanoparticles mainly through steric hindrance provided by its hydrated PEO chains. In contrast, choline-based ILs are attributed to the electrostatic repulsion during nanoprecipitation and particle assembly. These ionic interactions may influence the specific kinetics of the nucleation and growth of LNPs, which suggests that the ILs play a critical role in LNP formulation as a stabilizer. In addition, all IL-incorporated LNPs showed good monodispersity with low PDI values of less than 0.1, specifically, 0.02 ± 0.02 for Hex, 0.07 ± 0.05 for Asp and 0.04 ± 0.01 for Glu, whereas F127 exhibited a slightly higher PDI of 0.12 ± 0.02. The single peak of number profiles for each sample in figure 2c also confirms the good monodispersity of samples with a size (number mean) of 129 ± 32 nm for Hex, 104 ± 29 nm for Asp, 124 ± 34 nm for Glu and 49 ± 14 nm for F127. The intensity profiles for each sample in figure 2d show consistent results with a size (intensity mean) of 156 ± 35 nm for Hex, 138 ± 37 nm for Asp, 157 ± 40 nm for Glu and 89 ± 35 nm for F127. Zeta potential (ZP) measurements were further performed to characterize the surface charge of the IL-incorporated LNPs in water, as shown in figure 2e. All IL-incorporated LNPs exhibited positive surface charges of approximately +20 mV. In comparison, the F127 showed a slightly lower ZP of +16 ± 7 mV. This ZP increase upon IL incorporation in LNPs may be attributed to the permanently charged quaternary ammonium group of choline cation. Although choline is hydrophilic, it can localize at the lipid–water interface where it electrostatically interacts with anions (Hex, Asp and Glu) and lipid headgroups in LNPs, contributing to an electrostatic stabilization to prevent particle aggregation.

To examine the effect of IL incorporation on the pH-dependent protonation behaviour of LNPs, we employed the 6-(p-toluidino)-2-naphthalenesulfonic acid (TNS) fluorescence assay. TNS is a negatively charged solvatochromic dye that fluoresces more strongly in proximity to cationic environments, making it a useful probe for assessing surface charge and further deriving apparent pK_a of LNPs [41,42]. However, unlike conventional LNP systems containing only a single ionizable component (ionizable lipid), our IL-incorporated LNPs contain additional cationic species, i.e. the permanently charged choline moiety from the ILs. This

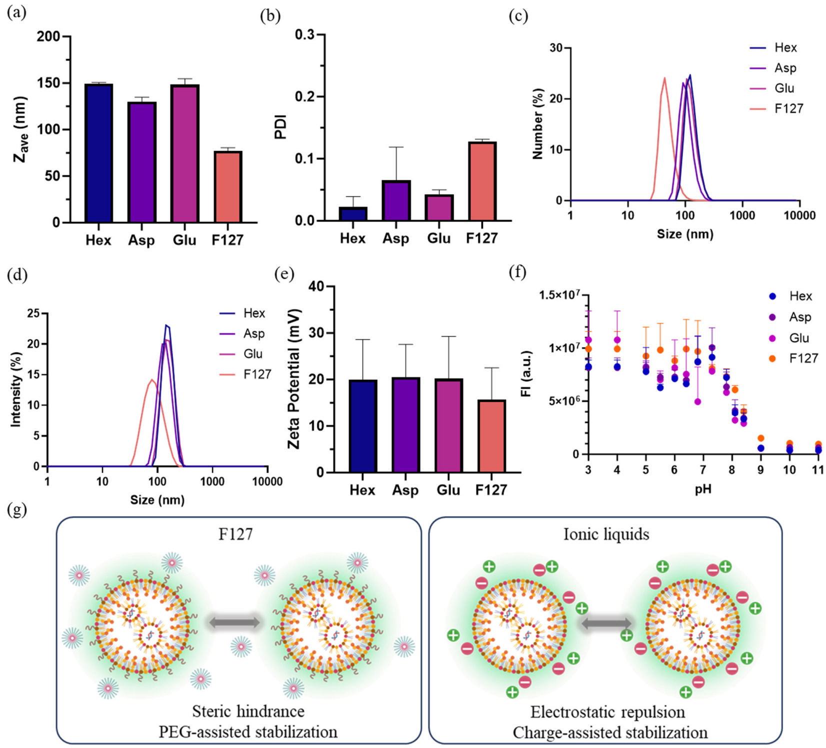

Figure 2. (a) Particle size (Z_{ave}), (b) PDI, (c) number profile, (d) intensity profile from DLS measurement and (e) ZP of LNPs stabilized by choline-based ILs (Hex, Asp and Glu) and F127 in water, $n = 3$. (f) TNS fluorescence intensity (FI) of LNPs measured as a function of pH, $n = 4$. (g) Stabilization of LNPs via the steric hindrance of copolymer Pluronic F-127 (left frame) versus the electrostatic repulsion of Cho Glu IL (right frame).

complexity prevents the determination of a discrete apparent pK_a for the ionizable lipid. Therefore, we measured TNS fluorescence intensity (FI) changes as a function of pH to qualitatively reveal the surface cationic density of LNPs.

The pH-dependent TNS FI of all samples is observed in figure 2f, revealing significantly higher FI at acidic conditions than basic conditions, which correlates with the protonation of SM-102 headgroups. Notably, at the pH range 5 to 7, the FI of all IL-incorporated formulations demonstrated was weaker compared to the F127 LNPs (with $p < 0.01$ for Hex versus F127, $p < 0.05$ for Asp versus F127 and Glu versus F127 at pH 5.5). This drop in FI can possibly be attributed to the deprotonation of the side-chain carboxyl group (p$K_a \approx 4.3$) [43] in ILs at this pH range. The negatively charged carboxyl groups can neutralize the protonated SM-102 headgroups via electrostatic interaction, reducing the effective surface charge experienced by the negatively charged TNS molecules. As a comparison, F127 wraps LNPs in a neutral PEO corona that sterically prevents aggregation, as sketched in figure 2g. A similar trend (weaker FI of IL-incorporated LNPs with $p < 0.0001$ comparing IL LNPs to F127 LNP) was also observed at the high pH conditions over 9, which suggests that the surface charge of IL-incorporated LNPs was neutralized more efficiently compared to F127 LNPs.

Colloidal stability of these IL-incorporated LNPs without traditional PEG-lipid or polymer stabilizers is another important property, as aggregation of LNPs can significantly impair their biodistribution, cellular interaction and efficiency. To assess the long-term stability of samples, we tested the particle size and PDI after 10 day and 28 day storage at room temperature, as shown in figure 3a,b. Over the first 10 days, all LNPs maintained stable particle size and reasonable monodispersity with PDI less than 0.15, although a clear PDI increase was observed compared to their initial values. After 28 day storage, there is still no obvious variations in both particle size and PDI for Asp and Glu. However, Hex LNPs showed a significant increase in

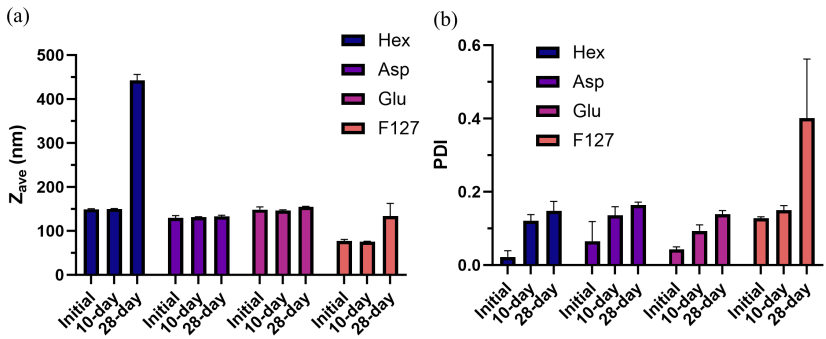

Figure 3. Stability of LNPs incorporating choline-based ILs (Hex, Asp and Glu) and F127 after 10 day and 28 day storage at room temperature. (a) Particle size (Z_{ave}) and (b) PDI, $n = 3$.

particle size to over 400 nm with a PDI of 0.15. This could be related to the limited polarity and greater hydrophobicity of hexanoate anion compared to amino acid anions with additional amino acid groups, as shown in figure 1. Thus, Hex probably partitions more into the lipid bilayer, contributing less to colloidal stability. In contrast, F127 showed poor stability after 28 day storage, with both particle size and PDI increasing to 134 ± 30 nm and 0.4 ± 0.2, respectively, which indicated obvious heterogeneity and particle aggregation. These stability results highlight the superior long-term stability of amino acid-based IL-incorporated LNPs with particle size of approximately 150 nm and good monodispersity throughout the 28 day period, confirming that the incorporation of choline-based ILs can stabilize LNPs without the need for other stabilizers. The good stability of Asp and Glu could be attributed to multiple effects, such as effective electrostatic repulsion, interfacial interactions with lipids and ion pairing between choline cations and amino acid anions of ILs. This finding also matches the previous reports that choline amino acid ILs can possess stable interactions with lipid bilayers and polymeric nanoparticles [24,40,44]. For comparison, the poor stability of Hex and F127 LNPs underscores the important role of the chemical structure of the IL anion and interfacial affinity in maintaining long-term colloidal stability, where insufficient hydration or weaker electrostatic screening may lead to gradual LNP growth or aggregation.

The good colloidal stability of these IL-incorporated LNPs makes them prospective nanocarriers for mRNA delivery. Enhanced green fluorescent protein (EGFP) mRNA (996 nt) was loaded using these LNPs at an N/P ratio of 6. The physicochemical properties of the EGFP mRNA-loaded LNPs were evaluated to determine the effect of IL inclusion on particle stability, uniformity, ZP and encapsulation efficiency (EE). As shown in figure 4a, the encapsulation of mRNA in LNPs led to an obvious increase in particle size across all formulations compared to their blank counterparts in figure 2a. Specifically, after mRNA encapsulation, the particle size Z_{ave} of Hex LNPs increased from 149 ± 1 nm to 481 ± 5 nm, Glu LNPs from 148 ± 6 nm to 329 ± 8 nm and Asp LNPs from 130 ± 5 nm to 283 ± 3 nm, whereas F127 LNPs increased from 77 ± 3 nm to 219 ± 1 nm. In addition, mRNA encapsulation also caused a noticeable increase in PDI values across all IL-incorporated samples (figure 4*b*), with Hex LNPs increasing from 0.02 ± 0.02 to 0.23 ± 0.03, Glu-LNPs from 0.04 ± 0.01 to 0.14 ± 0.06 and Asp-LNPs from 0.07 ± 0.05 to 0.16 ± 0.04. Despite these increases, all PDI values remained within a reasonable monodisperse range (PDI < 0.25), and the single dominant peak in both number-based profile (figure 4c) and intensity-based profile (figure 4d) for each sample further confirmed the size increase and uniformity of the mRNA@LNPs. This size increase of F127 LNPs upon mRNA encapsulation is consistent with our previous work [37]. The observed increase in particle size may result from the electrostatic interaction between the negatively charged phosphate backbone of mRNA and the ionizable lipid SM-102 during formulation, promoting nucleation and growth of larger lipid assemblies.

ZP measurements (figure 4e) revealed that all mRNA-loaded LNPs maintained positive surface charges, which are comparable to those of the blank LNPs in figure 2*e*. This indicates

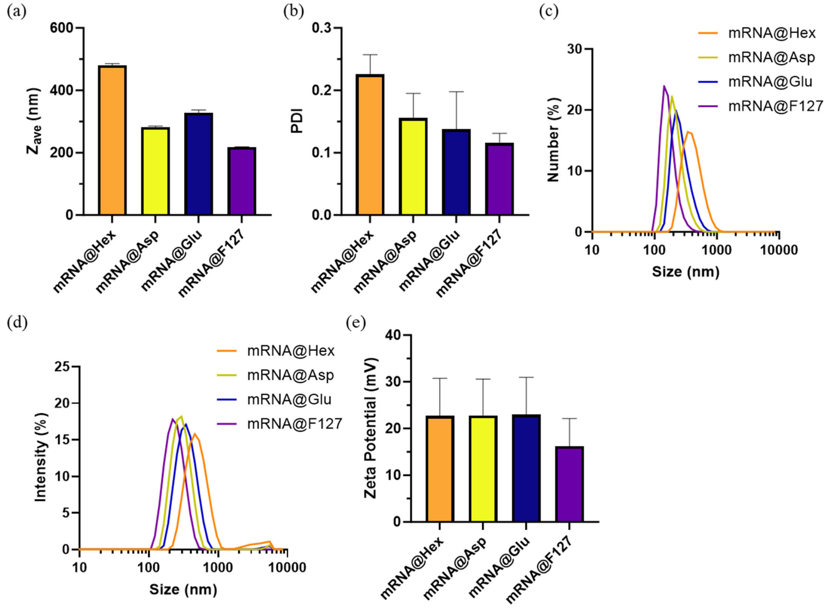

Figure 4. (a) Particle size (Z_{ave}), (b) PDI, (c) number profile, (d) intensity profile from DLS measurement, and (e) ZP of mRNA-loaded LNPs incorporating choline-based ILs (Hex, Asp and Glu) and F127 in water at room temperature, $n = 3$.

minimal effect of mRNA loading on surface charge as the ionizable lipid is in excess with respect to the aptamer in the LNPs with an N/P ratio of 6. ZP values of IL-incorporated mRNA@LNPs are larger than that of the F127 sample, consistent with the finding for the blank LNPs in figure 2d. This suggests localization of ILs at the LNP surface. Moreover, all IL-incorporated mRNA@LNPs exhibited a high EE, between 80% and 90%, as shown in electronic supplementary material, fig. S1, which is slightly higher than F127 (approx. 80%). This EE increase in the IL incorporation compared to F127 stabilization could result from the enhanced electrostatic interaction of IL-incorporated samples with the aptamer.

(b) *In vitro* mRNA transfection efficiency in macrophages

To evaluate the biological performance of the IL-incorporated LNPs for mRNA delivery, we employed the murine alveolar macrophage MH-S cell line, which serves as a representative *in vitro* model of pulmonary resident macrophages. Alveolar macrophages have been widely used in nanoparticle delivery research owing to their phagocytic activity and immunological responsiveness [38,39,45,46]. Fluorescence-activated cell sorting (FACS) analysis was performed to evaluate nano-bio interactions and mRNA transfection efficiency using Cy5-labelled EGFP mRNA. The mRNA-cell association tracked by the Cy5 fluorescent signal and mRNA transfection efficiency tracked by the signal of the EGFP mRNA-expressed green fluorescent protein (GFP) in the cytoplasm of the MH-S cells was assessed to determine the effect of IL incorporation on intracellular gene delivery of mRNA@LNPs.

As shown in figure 5a,b, MH-S cells treated by the four Cy5-EGFP mRNA@LNPs provided a high percentage of Cy5-positive cells, all over 90% with a Cy5 mean FI (MFI). This indicates the ability of IL-incorporated LNPs to efficiently deliver mRNA, comparable to the F127 LNPs. The cell viability (electronic supplementary material, fig. S2) after the 24 h treatment of F127 LNPs is approximately 70%, which is slightly lower than that (approx. 85%) after the 4 h treatment previously reported in [37], owing to the extended incubation time. The three IL-incorporated LNPs demonstrated approximately 85% cell viability, suggesting their improved biocompatibility compared to F127 LNPs. Notably, EGFP mRNA expression of cells

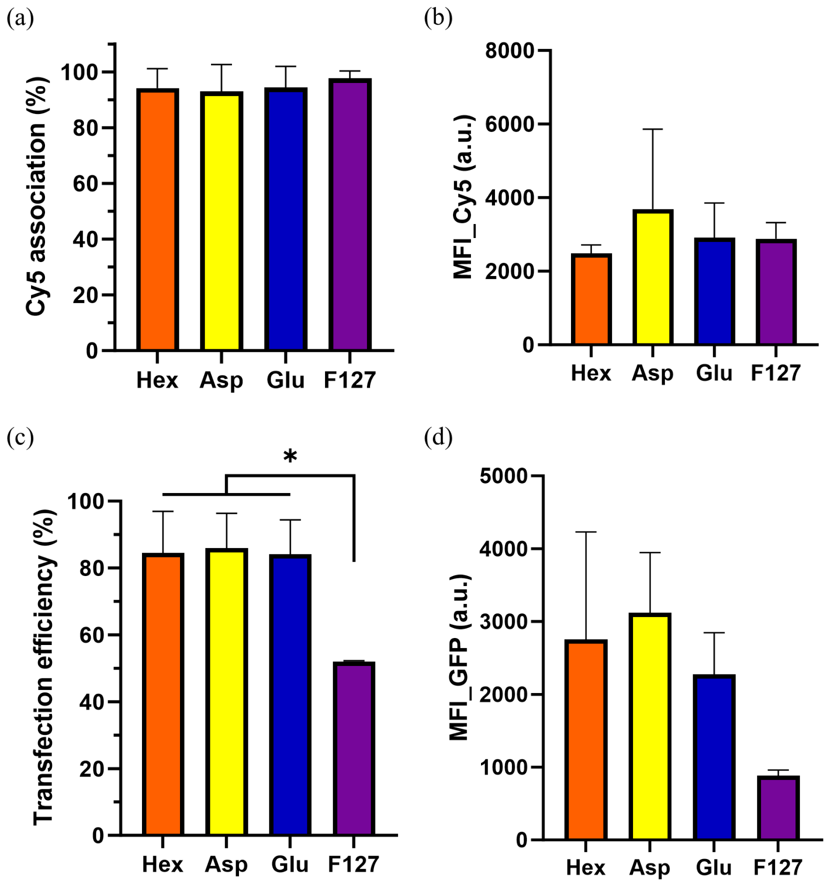

Figure 5. (a) mRNA-cell association and (b) mean fluorescence intensity (MFI) of Cy5 signal per cell, (c) mRNA transfection efficiency and (d) MFI_GFP of MH-S cells treated by Cy5-EGFP mRNA-loaded LNPs at a mRNA concentration of about 9 µg ml^{-1}, characterized using FACS, $n = 3$. The data in (c) were analysed by one-way ANOVA (unpaired) with Dunnett's post hoc test comparing IL LNPs to F127 LNP, *$p < 0.05$.

in figure 5c,d demonstrated that Hex LNPs (85 ± 11%), Asp LNPs (86 ± 9%) and Glu LNPs (85 ± 9%) achieved significantly higher transfection, in terms of the percentage of cells expressing the GFP, compared to F127 LNPs (52 ± 6%), $p < 0.05$. The GFP MFI values were highest for IL-incorporated LNPs, which are more than double the MFI compared to F127 LNPs. These findings reveal that, although all LNPs performed efficient cellular association of mRNA cargo, indicated by the similar Cy5 signal, the incorporation of ILs instead of copolymer F127 in LNPs obviously improved intracellular mRNA expression. This suggests that the transfection efficiency of LNPs is not only attributed to cell association, but also post-internalization processes such as endosomal escape. The discrepancy between the percentage of cells showing Cy5-positive signals (figure 5a) and percentage of cells having positive GFP expression (figure 5c) for F127-stabilized LNPs highlights that surface decoration with neutral polymers may limit endosomal escape, despite comparable cell uptake. For the IL-incorporated LNPs, the cationic choline-based ILs could play an important role in enhancing mRNA delivery efficacy, potentially facilitating intracellular delivery mechanisms including proton sponge effects [7,8,47] under endosomal acidification progress. Together, these findings validate the potential of choline-based ILs as bioactive components in LNPs, enhancing gene delivery to hard-to-transfect macrophages.

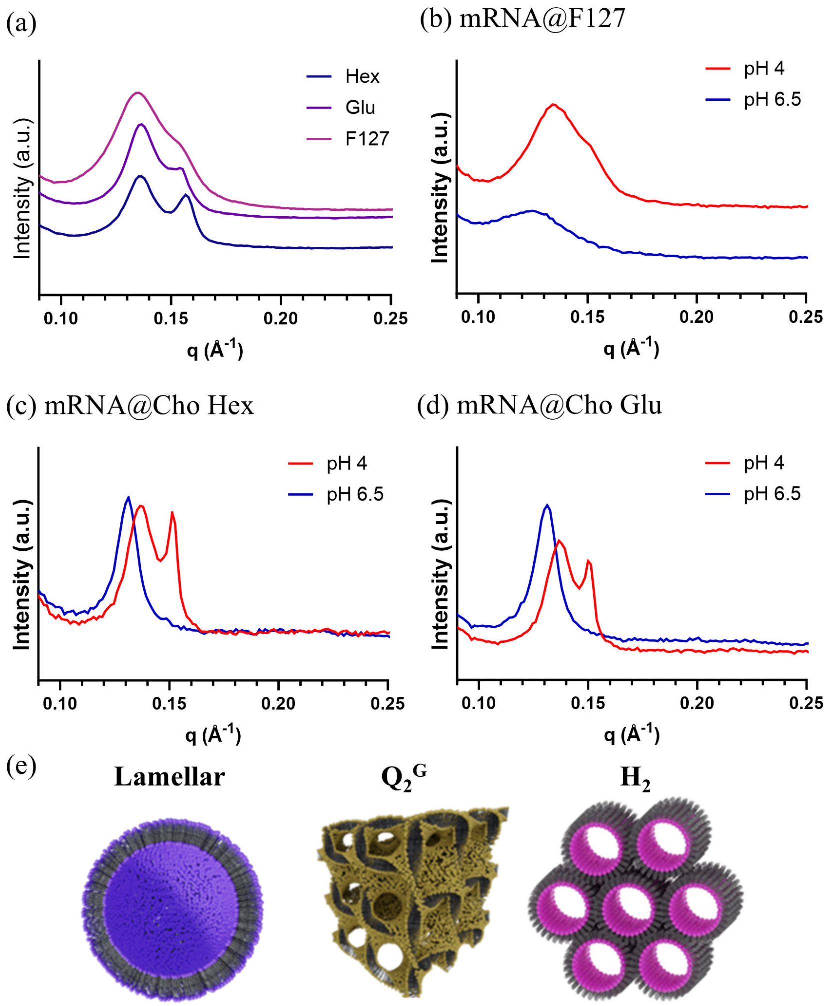

Figure 6. (a) Synchrotron SAXS patterns for mRNA-loaded LNPs with the incorporation of Cho Hex, Cho Glu and F127 in water (pH 5.6). (b–d) SAXS patterns of (b) mRNA@F127, (c) mRNA@Cho Hex and (d) mRNA@Cho Glu at pH 4 and 6.5. (e) Three-dimensional sketches of lamellar phase, gyroid cubic phase (Q_2^G) and hexagonal phase (H_2). Adapted with permission from [36].

(c) pH-dependent lipid mesophase evolution of ionic liquid-incorporated lipid nanoparticles

To better understand the superior mRNA transfection efficiency of those IL-incorporated mRNA@LNPs, we further examined the internal lyotropic liquid crystalline nanostructures of those mRNA@LNPs (Hex, Glu, and F127) in Milli-Q water (pH 5.6) and under different pH conditions (pH 4 and 6.5) using synchrotron small-angle X-ray scattering (SAXS), see figure 6. In this section, Cho Glu is selected as the representative choline amino acid IL; the similar properties and performances between Asp LNPs and Glu LNPs were previously discussed. The rationale for this pH-dependent structural characterization is based on the biological process of endosomal maturation, during which internalized LNPs encounter a progressively acidified environment from early endosomes (pH ~ 6.5) to late endosomes (pH ~ 5.5) and lysosomes (pH ~ 4.5). The important role of pH-dependent lyotropic liquid crystalline lipid mesophase transitions into the inverse phases such as inverse cubic or hexagonal phases during the endosomal escape and transfection efficiency of RNA-loaded LNPs has been reported in the

Table 2. Peak position and corresponding d-spacing of the first peak in SAXS plots of mRNA-loaded LNPs in figure 6.

sample	q of the first peak (Å^{-1})			d-spacing (Å)		
	water	pH 4	pH 6.5	water	pH 4	pH 6.5
mRNA@F127	0.135	0.134	0.125	46.4	46.9	50.3
mRNA@Cho Hex	0.137	0.137	0.131	46.0	46.0	47.9
mRNA@ Cho Glu	0.137	0.137	0.131	46.0	46.0	47.9

literature [36,37,48–50]. The F127 LNPs in water displayed a broader peak at 0.135 Å^{-1} and a weak shoulder observed at approximately 0.15 Å^{-1}, see figure 6a. In addition, mRNA@Hex and mRNA@Glu in water demonstrated two distinct peaks, with both the first peak at 0.137 Å^{-1} and the second peak at 0.157 Å^{-1} for Hex and 0.154 Å^{-1} for Glu. Subsequently, those samples were placed in different pH conditions (4 and 6.5), which replicate the acidification process during endosomal maturation following cellular uptake. As shown in figure 6*b–d*, at pH 6.5, mRNA@F127 showed a single broad peak at 0.125 Å^{-1} (corresponding to a d-spacing of 50.3 Å) and lacked a second peak. As pH reduces to 4.0, the peak position shifted to a larger q of 0.134 Å^{-1} (with a reduced d-spacing of 46.9 Å), which suggests a compaction of the internal nanostructure owing to a stronger interaction between the protonated ionizable lipid and mRNA at a lower pH condition. By contrast, mRNA@Hex and mRNA@Glu performed obvious structural transitions in response to pH change, see figure 6c,d. At pH 6.5, both formulations presented a single dominant peak at 0.131 Å^{-1} with a d-spacing of 48.0 Å, which is slightly smaller than that of mRNA@F127 (50.3 Å). Upon acidification to pH 4.0, two sharp peaks at 0.137 Å^{-1} and 0.150–0.151 Å^{-1} can be observed for both samples. A summary of all values of d-spacing is listed in table 2.

For all samples in water and pH 4, the first peak at approximately 0.135 Å^{-1} can be allocated to the lamellar structure of mRNA backbones sandwiched in lipid layers with a d-spacing of approximately 46.4 Å, which is in good agreement with the literature [37,51]. The position of the weak shoulder of mRNA@F127 and the second peaks of mRNA@Cho Hex and mRNA@Cho Glu at approximately 0.15 Å^{-1} matches the peak position of gyroid cubic Q_2^G phase (figure 6e) of blank F127 LNPs at pH 4, reported in our previous work [37]. Therefore, we speculate that the second peak at approximately 0.15 Å^{-1} can be associated with some presence of inverse cubic or hexagonal lipid mesophases at the lipid-domain regions, though we only observed a single peak in this work. For comparison, SAXS diffraction profiles of both blank Hex and Glu LNPs only displayed a single peak at 0.162 Å^{-1}, see electronic supplementary material, fig. S3. The increase of the second peak in IL-incorporated mRNA-loaded samples suggests that more ordered inverse lipid mesophases existed compared to mRNA@F127. Thus, the clearer and stronger second peaks in IL-incorporated systems may provide a mechanistic insight into their superior mRNA transfection observed in figure 5. This structure–function relationship underscores the role of choline-based ILs in facilitating non-lamellar phase evolution during the endosomal acidification.

3. Conclusion

This study demonstrates the successful development of IL-incorporated LNPs, in particular using choline amino acid ILs, as potent nanocarriers for mRNA transfection in macrophages. By replacing traditional polymeric stabilizers like F127 with three ILs, Cho Hex, Cho Asp and Cho Glu, we systematically investigated their physicochemical properties, including particle size, surface charge, long-term colloidal stability and pH-responsive lipid mesophase behaviour. Choline amino acid ILs (Cho Asp and Cho Glu) exhibited superior colloidal stability over 28 days than Cho Hex and F127. All three IL-incorporated mRNA@LNPs demonstrated efficient cellular association and enhanced mRNA transfection (approx. 85%) in MH-S macrophages compared to F127 controls. Combining SAXS and *in vitro* cellular results suggested the potential connection between pH-responsive nanostructure change within IL-incorporated LNPs and their superior gene expression. Overall, this work highlights the versatility of incorporating choline-based ILs, especially choline amino acid ILs, in LNP systems for mRNA delivery, providing a rational design strategy for nanocarriers in gene therapy.

4. Material and methods

(a) Materials

SM-102 (Heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate) with greater than 96% purity was purchased from Advanced Molecular Technologies (Scoresby, Victoria, Australia). Monoolein (MO, > 99%) was purchased from Nu-chek-Prep, Inc (Elysian, MN, USA). Choline hydroxide (46 wt% aqueous solution), choline bicarbonate (80 wt% aqueous solution), hexanoic acid (≥ 99%), L-glutamic acid (≥ 99%) and L-aspartic acid (> 98%), Pluronic F127 (F127), cholesterol, 2-(*p*-toluidino)-6-naphthalene sulfonic acid (TNS), citric acid (≥ 99.5%), sodium citrate monobasic (≥ 99.5%), sodium phosphate dibasic (≥ 99.0%), sodium phosphate monobasic (≥ 99.0%) and sodium bicarbonate (≥ 99.7%) were purchased from Sigma-Aldrich (St. Louis, USA). Methanol (analytical reagent) was obtained from Univar Ajax Finechem (NSW, Australia). Cy5-labelled N1-methylpseudouridine-based modified EGFP was purchased from BASE mRNA facility in The University of Queensland, Australia. NucBlue® Live ReadyProbes® Reagent, RPMI + GlutaMAX (1×) medium, Opti-MEM (1×) reduced serum medium, Quant-iT RiboGreen RNA Kit was purchased from Thermo Fisher Scientific (Waltham, MA, USA). Hemacolor® Rapid Staining kit was purchased from Sigma-Aldrich (St. Louis, USA). All chemicals were used without further purification.

(b) Ionic liquid synthesis

Three ILs, choline hexanoate (Cho Hex, 1 : 1 molar ratio), choline glutamate (Cho Glu) and choline aspartate (Cho Asp), were synthesized following previously reported methods [40,52]. For Cho Hex, equimolar amounts of hexanoic acid and choline bicarbonate were mixed, with the temperature controlled below 10°C initially. The reaction mixture was then stirred continuously at room temperature for 24 h to achieve complete neutralization. For Cho Glu and Cho Asp, equimolar amounts of the respective amino acids (L-glutamic acid or L-aspartic acid) and choline hydroxide solution (46 wt% in water) were combined under continuous stirring at 70°C. To facilitate complete solubilization of the amino acids, an additional 8 ml of water was added. These reactions were maintained at 70°C with stirring for 24 h. Subsequently, excess water from all synthesized ILs was removed using rotary evaporation under reduced pressure, followed by freeze-drying to yield the final ILs. The chemical structures and purities of the resulting ILs were confirmed by ^{1}H NMR spectroscopy. Final water content, determined by Karl Fischer titration, was consistently below 1 wt% for all ILs.

(c) Lipid nanoparticle formulation

Lipids with SM-102/MO/cholesterol molar ratio of 50/30/20 were dissolved in methanol to achieve a total concentration of 15 mg ml^{-1}. Citrate buffer (10 mM, pH 3) and 0.5 mg ml^{-1} IL (or F127) without and with mRNA for blank LNPs and mRNA@LNPs was used as the aqueous phase. Blank LNPs were prepared using the microfluidic system NanoAssemblr Benchtop platform (Precision NanoSystems, Vancouver, CA) with a total flow rate of 12 ml min^{-1} and flow rate ratio (solvent/aqueous) of 1 : 3. For mRNA@LNPs, samples were prepared using the NanoAssemblr Spark (Precision NanoSystems, Vancouver, CA). Formulated samples were dialysed in Milli-Q water for 2 h at room temperature to remove the solvent residue. The efficiency of mRNA encapsulation was assessed using the Quanti-iT RiboGreen assay (Thermofisher) as per the manufacturer's protocol.

(d) Lipid nanoparticle characterization

(i) Dynamic light scattering measurement

The size and PDI of LNPs (approx. 0.5 mg ml^{-1} in Milli-Q water) were measured using a Zetasizer APS (Malvern Panalytical, UK) and analysed using refractive indices (1.33 for water and 1.45 for lipids); mRNA-loaded LNPs were tested using the Zetasizer Nano ZS (Malvern Panalytical, UK). Zeta potential measurements were performed on LNPs (approx. 0.1 mg ml^{-1} in Milli-Q water) and measured using a Malvern Zetasizer Nano ZS (Malvern Instruments, UK) with polystyrene ZP cells (Malvern).

(ii) 6-(p-toluidino)-2-naphthalenesulfonic acid fluorescence assay

The surface cationic property of the LNPs was evaluated using the TNS binding assay [51,53,54]. Specifically, a 150 µM stock solution of TNS was prepared in DMSO, and 2 µl of this solution was added to a total reaction volume of 150 µl. The reaction mixture included 115 µl of Milli-Q water, 20 µl of LNPs and 15 µl; 100 mM pH-specific buffers. Citric acid buffers (pH 3–6), sodium phosphate buffers (pH 6.5–8.5) and sodium carbonate buffers (pH 9–11) were used. FI was measured at an excitation wavelength of 325 nm and an emission wavelength of 435 nm using a PerkinElmer Multimode EnSight plate reader.

(iii) Synchrotron small-angle X-ray scattering

Synchrotron SAXS experiments were performed at the SAXSWAXS beamline of the Australian Synchrotron (ANSTO) and data were analysed as previously described [36,54]. Original LNP samples at a concentration of 3.75 mg ml^{-1} were concentrated to approximately 15 mg ml^{-1} using centrifugal filtration. The samples were then diluted at a 1 : 1spa volume ratio with buffer solutions (100 mM pH 4 citrate buffer or 100 mM pH 6.5 sodium phosphate buffer) to achieve a final buffer concentration of 50 mM. Samples were loaded into a UV-clear 96-well plate and measured with a 2 s exposure time.

(e) *In vitro* cellular experiments

The murine alveolar macrophage cell line MH-S was cultured in RPMI medium supplemented with 10% FBS, 2 mM 2-mercaptoethanol and 5% penicillin–streptomycin. MH-S cells were seeded in 96-well plates at a density of 3×10^4 cells per well and incubated overnight in their respective standard media at 37 °C. After centrifugation at 400×*g* for 5 min, the media were removed and 10 µl of EGFP mRNA-loaded LNPs was added to 200 µl of Opti-MEM (1×) in each well. Cells were incubated for 24 h in a light-protected environment. Following incubation, plates were centrifuged, and cells were detached using TrypLE Express and resuspended in flow cytometry (FACS) buffer containing DAPI for viability analysis. FACS was performed using a LSRFortessa X-20 system (BD Biosciences, USA). Viability was assessed based on DAPI fluorescence, mRNA association was assessed by Cy5 fluorescence and transfection efficiency was evaluated via GFP expression.

Data accessibility. Supplementary material is available online [55].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. H.Y.: conceptualization, data curation, formal analysis, investigation, methodology, writing—original draft; N.M.: data curation, formal analysis, investigation, methodology, writing—review and editing; Q.H.: data curation, formal analysis, investigation, methodology, writing—review and editing; M.E.M.: investigation, methodology, writing—review and editing; B.D.: data curation, formal analysis, writing—review and editing; S.B.: data curation, funding acquisition, resources, supervision, writing—review and editing; L.v.H.: data curation, funding acquisition, writing—review and editing; C.D.: data curation, formal analysis, funding acquisition, resources, supervision, writing—review and editing; J.Z.: conceptualization, data curation, funding acquisition, resources, supervision, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

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